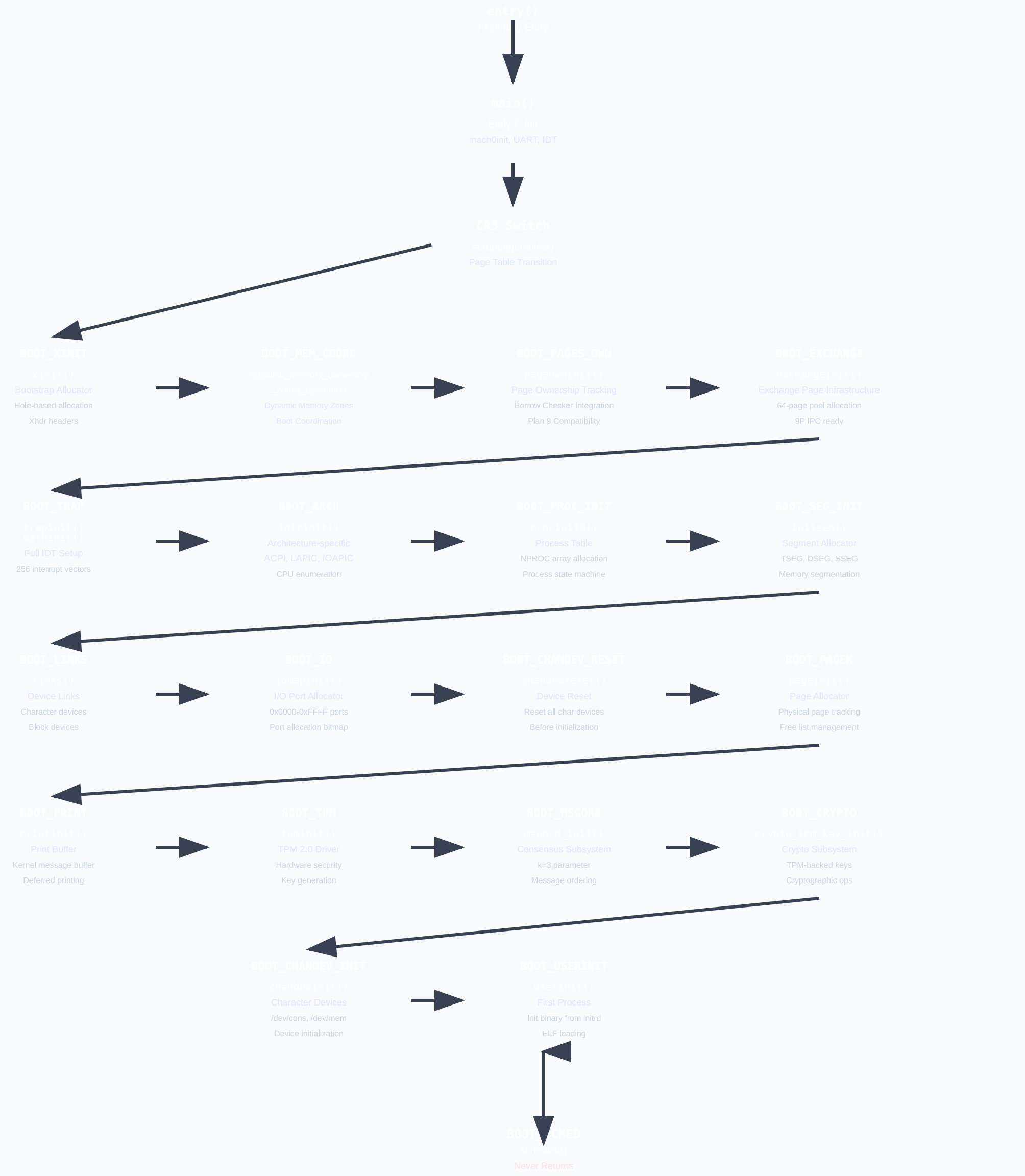


Lux9 Boot State Machine

20-State Sequential Boot Process (Actual Implementation)



Error Handling

Each state includes comprehensive error checking
Any failure results in immediate kernel panic
Boot log provides detailed state tracking via set_boot_state()
State validation prevents invalid transitions

Critical Dependencies

CR3 Switch → Memory Allocation | Console Ready → print() | Timer Setup → schedinit() | Security Hardware → Crypto
Each transition validates prerequisites to prevent undefined behavior during critical boot sequence